

JAPAN ELECTRONIC COMPONENTS/SEMICONDUCTORS

Earnings summary: we expect a series of upward revisions in the next reporting season; maintain Buy on Ibiden/Murata/Taiyo Yuden/Renesas/Rohm/TDK, etc

Overall view: The earnings reporting season for our coverage universe is mostly completed as of mid-August. Overall, earnings were generally strong, and we revised up our previous estimates for 15 of the 19 companies that have announced results. Furthermore, we expect previous guidance to be beaten in the next reporting season, potentially leading to a series of upward revisions to full-year guidance. While investor reaction immediately following the results appears to be a “wait-and-see” approach, we expect the sector’s share prices to regain upward momentum toward the second half of the year, backed by strong fundamentals. Note that the gap in earnings between individual companies can be mostly explained by (1) the strength/weakness of benefits from AI/DCs, and (2) the strength/weakness of the impact from material/cost factors. We believe the direction is in line with our pre-earnings stock classifications.

Individual stocks: We summarize the catalysts and our views for major stocks ahead of (and around) the next reporting season. **(1) Ibiden (Buy):** We expect an upward revision to its medium-term plan and additional capacity expansion. **(2) Murata Mfg. (Buy, on CL):** We see upside to 2Q guidance and expect another upward revision to full-year guidance. For **Taiyo Yuden (Buy)**, we expect a rapid earnings recovery from 2Q onward. For both companies, we expect the emergence of mix effects driven by expansion for AI/DCs and price hike effects in 2H. **(3) Renesas (Buy):** We expect qoq earnings expansion and room for margin improvement in FY12/27. **(4) Rohm (Buy) / TDK (Buy):** We expect significant upward revisions vs. initial guidance for both companies. **(5) Other Buy-rated stocks:** We expect **Kyocera (Buy)** to revise full-year guidance upwards again and raise its medium-term ROE target. We expect **MinebeaMitsumi (Buy)** to beat FY3/27 earnings expectations and see increased visibility on its medium- to long-term plan. For **Nidec (Buy)**, we highlight the status of the filing of its securities report, due on September 30, as the most important focus point. **(6) Neutral-rated stocks:** We expect upward revisions for **Nichicon, Hirose Electric, and NGK**.

See below for details. Note that we have raised our earnings estimates and target price for Rohm (see [LINK](#)).

Daiki Takayama
+81(3)4587-9870 |
daiki.takayama@gs.com
Goldman Sachs Japan Co., Ltd.

Mitsuhiro Icho
+81(3)4587-9836 |
mitsuhiro.x.icho@gs.com
Goldman Sachs Japan Co., Ltd.

Makoto Takahara
+81(3)4587-4270 |
makoto.takahara@gs.com
Goldman Sachs Japan Co., Ltd.

Yuji Hidaka
+81(3)4587-3656 | yuji.hidaka@gs.com
Goldman Sachs Japan Co., Ltd.

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

Earnings summary of major stocks: earnings points and next catalysts

Ibiden (Buy): We expect the next catalysts to be an upward revision to the medium-term plan and additional capacity expansion

(1) Electronics division OPM: The full-year electronics OPM guidance was revised up from 22.7% previously to 29.3%. According to the company, this is mainly due to selling prices rather than product mix.

(2) For CPUs: We estimate OPM for the largest CPU customer was just under 20% in 1Q, and we see the potential for this to rise to nearly 30% heading into 4Q.

(3) Medium-term plan: Given the higher-than-expected rise in profitability in FY3/27, we expect the medium-term plan (operating profits of ¥150 bn in FY3/28 and ¥300 bn or more in FY3/31) could be revised up at the 1H (2Q) results.

(4) Additional capacity expansion: the timing of the announcement is a matter of negotiation with customers and it is unclear whether it will be in 2Q or 3Q, but the company wants to decide during FY3/27; it is expected to take about four years from building construction to the start of production (even if decided in 2H this year, production contribution will be in late FY3/30)

Murata Mfg. (Buy, on CL): We expect the next catalysts to be 2Q earnings upside and a further upward revision to full-year guidance. In 2H, we expect further mix effects and price effects.

(1) 2Q assumptions are conservative: Given the MLCC book-to-bill (BB) ratio of 1.47x in 1Q, we believe the company's assumptions for 2Q MLCC sales (up 2% qoq) are still conservative (our 2Q estimates are higher than the company's assumptions).

(2) Order backlog: While lead times are tending to lengthen compared to before, we estimate that the majority are still under three months. We do not believe it will take much time for these to be recognized as sales.

(3) Soundness: The company believes that, excluding some orders for distributors, orders are sound and backed by actual demand.

(4) MLCCs for AI/DCs: The annual sales growth rate was revised up from +85-90% previously to +100-105%. Since sales for AI/DCs are trending stronger heading into 2H, we see the potential for a greater-than-expected mix improvement (higher marginal profit ratio) effect.

(5) Conditions for price hikes: The three conditions for price hikes were cited as (A) cost increases in the supply chain, (B) the gap between peers' price hikes and market prices, and the emergence of excess replacement demand for the company's products, and (C) channel inventory management (restraining hoarding by distributors). Management's stance has gradually shifted from not considering price hikes to considering partial price hikes, and we recognize that the conditions for proceeding with partial price hikes are falling into place.

Taiyo Yuden (Buy): We expect the next catalyst to be a transition to a rapid earnings recovery phase in 2Q/3Q

(1) MLCC orders/sales expanding rapidly: Given the MLCC BB ratio of 1.72x, our

impression is that 2Q MLCC sales (up 15-19% qoq) are reasonable (there was a sales shortfall of about ¥4.0 bn in 1Q due to temporary factors, and the low 1Q base is also a reason for the high qoq growth rate in 2Q).

(2) 2Q is a turning point for rapid earnings recovery: With MLCC prices seeing almost no qoq decline in 2Q, utilization rates are rising from 85% in 1Q to 95% in 2Q. We believe strong marginal profit ratio effects will emerge, entering a phase of rapid earnings improvement.

(3) 2H: In 2H, we believe there is potential for further mix improvement due to a higher sales composition for AI/DCs, as well as some price hike effects. Sales for AI servers were also revised up from +80-85% previously to +150%. We believe the company will also proceed with some inventory buildup, entering a phase where positive operating leverage is even more likely to emerge.

(4) FY3/28: If a high MLCC sales growth rate can be maintained (our estimates are +26% yoy in FY3/27 and +25% yoy in FY3/28), our impression is that a high marginal profit ratio will contribute, making yoy profit growth a straightforward calculation.

Renesas (Buy): We expect near-term qoq earnings expansion and profitability improvement in FY12/27 due to in-house capacity expansion to be the catalysts

(1) Rapid increase for DCs: Sales for data centers (DCs), which account for about 85-90% of the infrastructure segment, were slightly over ¥160 bn in FY12/25. We believe they are on pace to exceed ¥300 bn (up 85-90% yoy) in FY12/26.

(2) Breakdown of DC sales: Within DC sales, our impression is that digital power (PMIC) is doubling yoy, while memory interfaces and others are increasing at a pace of over 50% yoy.

(3) PMIC: The company infers that it is the sole supplier for one ASIC company, which has contributed significantly to the qoq increase in CY2Q. We expect sales for ASICs to remain at a high level from 3Q onward, while sales for GPUs are trending stronger for new models.

(4) Overall company OPM: We expect a gradual qoq improvement in overall company OPM in FY12/26. On the other hand, in FY12/27, as in-house production increases and production volume grows (from around mid-FY12/27 following the ramp-up), we believe the pace of margin improvement could accelerate.

(5) Cost pass-through: The company's view is to first proceed with passing on material costs (the effects are not factored into the Jul-Sep guidance). Overall, the company's stance of catching up with market price changes remains unchanged.

Rohm (Buy): We assume the next catalyst is the possibility of a significant full-year upward revision

(1) Overall company earnings: The company notes the potential for highly leveraged earnings growth from 2Q onward. Specifically, 2Q earnings could be boosted by (A) qoq sales growth backed by the 1Q BB ratio of 1.2x, (B) rising utilization rates (55% in April-May, 65% in June, heading toward 70% or more in September), and (C) improved SiC yields (the temporary deterioration factor in 1Q has been resolved). Furthermore, effects from structural reforms and price revisions are expected in 2H. Having

re-examined these positive changes, we have upwardly revised our earnings estimates and target price (see [LINK](#)).

(2) SiC: Demand for both AI/DCs and EVs is strong, and the company is reportedly considering the need to restart idle equipment. During a very busy 1Q, the company had to use poor quality epi-wafers, which led to a deterioration in yields (a negative impact of ¥3.0 bn). Procurement has now been secured and the issue resolved. The company expects profitability to improve from 2Q.

(3) For AI/DCs: FY3/27 sales guidance was raised from ¥25.0 bn previously to ¥30.0 bn. Demand is reportedly particularly strong for Si-MOS used in 48V DC/DC converters and the like.

(4) Price hikes: Given the favorable pricing for AI/DCs, the company plans to actively proceed with price revisions for low-profitability applications and products. We expect price effects that exceed the pass-through of costs for materials such as precious metals to emerge heading into 2H.

TDK (Buy): we assume the next catalyst is the possibility of a significant full-year upward revision

(1) Overall company earnings: Our impression is that 1Q results and the outlook for all businesses are above expectations. We believe 2Q operating profits could exceed ¥100 bn (already factored into our estimates).

(2) Rising pack ratio for small rechargeable batteries: While a 7% yoy volume decline is planned due to reduced demand for smartphones and PCs, the rising pack ratio is absorbing the volume decline and becoming an additional profit growth driver. Cases of receiving orders for battery packs rather than cells are increasing not only for North American smartphones but also for Chinese high-end smartphones. Our impression is that rising ASPs are contributing more positively than expected (our impression is that profit margins remain unchanged even for battery packs rather than cells).

(3) Industrial batteries: Annual sales guidance was revised up from roughly ¥110 bn to roughly ¥150 bn. Most of the increase is for UPS batteries for AI/DCs, and our impression is that the acquired Malaysian startup (Liner Power) will begin contributing immediately.

(4) HDD-related: We believe profit margins could improve rapidly going forward, given that (A) rising utilization rates in the front-end process (wafer input) for HDD heads will contribute significantly to margin improvement, and (B) the company is able to achieve favorable pricing for both HDD heads and suspensions.

(5) Passive components: As a result of profitability management by BU (Business Unit) and ROIC-based management, the growth potential and profitability of the passive components business are increasing. In addition to the strengthening of MLCCs and aluminum electrolytic capacitors, thin-film inductors are expected to contribute to earnings going forward.

Other stocks for which we expect positive news flow going forward are as follows

Kyocera (Buy): we expect news flow within the year regarding further upside to FY3/27 guidance (a second upward revision) driven mainly by various AI/DC-related

components, and the possibility of an upward revision to the ROE target (8%) in the next medium-term plan (FY3/29-FY3/31) (note that management has already commented that the current medium-term target of 5% ROE by the final year in FY3/28 is scheduled to be achieved in FY3/27)

MinebeaMitsumi (Buy): the next focus points toward year-end are the upside potential for FY3/27 guidance driven mainly by bearings and motors, and increased visibility on achieving the medium-term plan for FY3/28/FY3/29

Nidec (Buy): the next major focus point is the status of the filing of the securities report due September 30

Nichicon (Neutral): we expect the possibility of an upward revision to FY3/27 guidance and an updated outlook for aluminum electrolytic capacitors for AI/DCs

Hirose Electric (Neutral): we expect further upside to FY3/27 guidance (a second upward revision) driven mainly by expansion for industrial applications

NGK (Neutral): we expect the possibility of an upward revision to FY3/27 guidance due to upward momentum in automotive ceramics earnings and expanded profit growth for electronics-related products in 2H

Price Target Risks and Methodologies

Exhibit 1: Valuation methodologies

Company Name	Ticker	Analyst	Rating	Price (As of 8/13)	Target Price	Upside/Downside	TP Methodology
Ibiden	4062.T	Daiki Takayama	Buy	21,560	26,700	24%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying +60% premium to our sector average EV/DACF multiple of 10X.
NGK Corp.	5333.T	Daiki Takayama	Neutral	6,006	6,600	10%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying -10% discount to our sector average EV/DACF multiple of 10X.
Niterra	5334.T	Daiki Takayama	Neutral	9,703	10,900	12%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying -10% discount to our sector average EV/DACF multiple of 10X.
MinebeaMitsumi Inc.	6479.T	Daiki Takayama	Buy	3,801	5,500	45%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, in-line with our sector average EV/DACF multiple of 10X.
Mabuchi Motor	6592.T	Daiki Takayama	Sell	1,697	1,400	-18%	Our target price is based on FY12/28E EV/GCI vs. CROCI/WACC, applying -30% discount to our sector average EV/DACF multiple of 10X.
Nidec	6594.T	Daiki Takayama	Buy	2,778	3,200	15%	Our target price is based on the average of FY3/27E-FY3/28E EV/GCI vs. CROCI/WACC, applying +30% premium to our sector average EV/DACF multiple of 8X.
Renesas Electronics	6723.T	Daiki Takayama	Buy	3,754	5,500	47%	Our target price is based on FY12/28E EV/GCI vs. CROCI/WACC, applying EV/DACF multiple of 13X.
TDK	6762.T	Daiki Takayama	Buy	3,235	4,900	51%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying +10% premium to our sector average EV/DACF multiple of 10X.
Alps Alpine	6770.T	Daiki Takayama	Neutral	2,208	2,200	0%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying -60% discount to our sector average EV/DACF multiple of 10X.
Hirose Electric	6806.T	Daiki Takayama	Neutral	27,715	30,400	10%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, in-line with our sector average EV/DACF multiple of 10X.
Japan Aviation Electronics Industry	6807.T	Daiki Takayama	Sell	2,335	2,000	-14%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying -60% discount to our sector average EV/DACF multiple of 10X.
Rohm	6963.T	Daiki Takayama	Buy	4,886	7,200	47%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying +20% premium to our sector average EV/DACF multiple of 10X.
Kyocera	6971.T	Daiki Takayama	Buy	3,905	4,300	10%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, in-line with our sector average EV/DACF multiple of 10X.
Taiyo Yuden	6976.T	Daiki Takayama	Buy	10,935	21,300	95%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying +80% premium to our sector average EV/DACF multiple of 10X.
Murata Mfg.	6981.T	Daiki Takayama	Buy*	8,311	12,900	55%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying +80% premium to our sector average EV/DACF multiple of 10X.
Nitto Denko	6988.T	Daiki Takayama	Neutral	3,551	3,200	-10%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, applying -40% discount to our sector average EV/DACF multiple of 10X.
Nichicon	6996.T	Daiki Takayama	Neutral	3,035	4,100	35%	Our target price is based on FY3/29E EV/GCI vs. CROCI/WACC, in-line with our sector average EV/DACF multiple of 10X.
MARUWA	5344.T	Mitsuhiro Icho	Buy	62,170	89,000	43%	Our target price is based on the correlation between EV/EBITDA multiple and EBITDA margin. (FY3/29E)
Maxell Ltd.	6810.T	Mitsuhiro Icho	Sell	2,397	1,900	-21%	Our target price is based on a P/B of 0.7X (FY3/29E)
IRISO Electronics	6908.T	Mitsuhiro Icho	Neutral	2,788	3,400	22%	Our target price is based on FY3/28E EV/GCI vs. CROCI/WACC, applying -30% discount to our sector average EV/DACF multiple of 8X.
Nippon Ceramic	6929.T	Mitsuhiro Icho	Sell	3,750	3,600	-4%	Our target price is based on FY12/28E EV/GCI vs. CROCI/WACC, applying -20% discount to our sector average EV/DACF multiple of 10X.
Kohoku Kogyo	6524.T	Mitsuhiro Icho	Neutral	4,840	6,000	24%	Our target price is calculated using a 10-year DCF model (terminal growth rate of 0%)

*On the APAC Conviction List. Target price is on a 12-month basis.

Source: Goldman Sachs Global Investment Research, LSEG Data & Analytics

Exhibit 2: Risks

Ticker	Company Name	Key Risks
6770	Alps Alpine	Key risks include forex fluctuations and changes in auto production.
6929	Nippon Ceramic	Greater-than-expected increase in automobile production, increase in the number of ultrasonic sensors installed per vehicle, emergence of sales growth for current sensors, additional share buybacks, and forex swings.
6908	Iriso Electronics	Changes in automobile production volume, changes in materials and distribution costs, changes in plant operations, changes in the competitive environment in growth fields like connectors for automotive powertrains, forex swings.
6594	Nidec	Slower-than-expected recovery in auto production volume, a greater-than-expected slowdown in the European economy, tougher competition in the water cooling module business, a further rise in materials prices.
6479	MinebeaMitsumi	Sharper-than-expected rises in input costs, a decline in auto production, and yen appreciation.
6807	Japan Aviation Electronics Industry	Increased interest in aerospace themes, strengthening of the relationship with Kyocera, upside in smartphone and automotive production, declining material costs.
6996	Nichicon	Exchange rate fluctuations, changes in demand, material costs, and power costs for aluminum electrolytic capacitors, fluctuations in EV market volumes, and the impact of subsidies.
6988	Nitto Denko	Material cost fluctuations, exchange rate fluctuations, changes in shipment volumes of smartphones, PCs, and tablets primarily in North America.
6762	TDK	Decline in smartphone production volume, higher input costs, and yen appreciation.
6806	Hirose Electric	Greater-than-expected yen appreciation/depreciation, changes in smartphone and auto production.
5334	Niterra	Changes in auto production, input cost fluctuations, the speed of new business development, and forex swings.
4062	Ibiden	Road map changes of main customers, weaker PC and server demand, and a reduction in diesel vehicle sales
6976	Taiyo Yuden	Weaker-than-expected smartphone demand, deterioration in MLCC supply/demand, and yen appreciation.
6592	Mabuchi Motor	Stronger-than-expected auto production volume, a decline in input costs, and yen depreciation.
5344	MARUWA	A decline in investment in applications that support end demand, such as AI/general-purpose servers, xEVs, and semiconductor manufacturing equipment; a decline in end product demand due to supply chain disruptions and inventory adjustments; the emergence of substitute technologies.
6981	Murata Mfg.	A decline in smartphone production volume, deterioration in MLCC supply/demand, and yen appreciation.
5333	NGK Corp.	Forex fluctuation, auto production volatility, and pace of the recovery in the semiconductor cycle.
6971	Kyocera	Slower-than-expected progress on planned management changes, inefficient investment or deployment of surplus capital, a delayed cycle recovery, and yen appreciation.
6524	Kohoku Kogyo	Optical components and devices business: Demand fluctuation for major end-use products (automobiles, telecommunications equipment such as communication stations, industrial machinery, home appliances, etc.) Lead terminals business: Sudden changes in the schedule of telecom submarine cable laying projects, reactionary decline in demand due to excessive component production for telecom submarine cable laying companies. Company-wide: FX fluctuations, political instability in major production locations such as Sri Lanka and Dongguan in China
6810	Maxell, Ltd.	A greater-than-expected recovery in automobile output, increased adoption of the company's automotive products, acceleration in margin improvement for consumer electronics products, lower costs relating to projector business downsizing, and yen depreciation.
6963	Rohm	Further increases in precious metal prices, delays in business portfolio transformation, uncertainty surrounding thereorganization and integration schemes, fluctuations in the market valuation of Toshiba shares.
6723	Renesas Electronics	A slowdown or delayed recovery in consumer electronics-related demand, stagnant automobile production, protracted reduction in semiconductor inventories, delayed recovery for industrial equipment applications, and yen appreciation

Source: Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Daiki Takayama, Mitsuhiro Ichio, Makoto Takahara and Yuji Hidaka, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Daiki Takayama Goldman Sachs Japan Co., Ltd., Mitsuhiro Ichio Goldman Sachs Japan Co., Ltd., Makoto Takahara Goldman Sachs Japan Co., Ltd., Yuji Hidaka Goldman Sachs Japan Co., Ltd..

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Ibiden, Murata Mfg., Renesas Electronics, Rohm, TDK and Taiyo Yuden is/are relative to the other companies in its/their coverage universe: Alps Alpine, Dai Nippon Printing, Hirose Electric, IRISO Electronics, Ibiden, Japan Aviation Electronics Industry, Kohoku Kogyo, Kyocera, MARUWA, Mabuchi Motor, Maxell Ltd., MinebeaMitsumi Inc., Murata Mfg., NGK Corp., Nichicon, Nidec, Nippon Ceramic, Niterra, Nitto Denko, Renesas Electronics, Rohm, TDK, TOPPAN Holdings, Taiyo Yuden

Company-specific regulatory disclosures

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	59%	43%

As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

Price target and rating history chart(s)

Compendium report: please see disclosures at <https://www.gs.com/research/hedge.html>. Disclosures applicable to the companies included in this compendium can be found in the latest relevant published research

Target price history table(s)**Taiyo Yuden (6976.T)**

Date of report	Target price (¥)	Closing price (¥)
05-Aug-26	21,300	11,165
08-Jun-26	21,200	14,975
08-May-26	7,100	6,663
23-Mar-26	4,900	3,926
12-Jan-26	4,300	3,405
02-Oct-25	3,800	3,382
05-Aug-25	3,200	2,841
07-Jul-25	3,100	2,488
09-May-25	3,200	2,202
31-Mar-25	3,400	2,467
07-Nov-24	3,600	2,781
01-Oct-24	4,100	3,081
02-Jul-24	4,800	4,105
08-May-24	4,000	3,643
07-Feb-24	4,100	3,431
06-Dec-23	4,500	3,627
07-Nov-23	4,400	3,695
04-Oct-23	4,800	3,963

Renesas Electronics (6723.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	5,500	3,442
08-Jun-26	5,400	4,139
24-Apr-26	3,800	3,141
23-Mar-26	3,600	2,276
05-Feb-26	3,300	2,767
12-Jan-26	2,900	2,380
24-Apr-25	2,500	1,670
31-Mar-25	2,900	1,988
06-Feb-25	2,700	2,306
31-Oct-24	2,600	2,110
01-Oct-24	2,900	2,161
25-Jul-24	3,400	2,584
02-Jul-24	3,800	3,014
03-Apr-24	3,400	2,650
08-Feb-24	3,200	2,542
27-Nov-23	3,300	2,563
26-Oct-23	2,900	2,106
04-Oct-23	3,000	2,148

Rohm (6963.T)

Date of report	Target price (¥)	Closing price (¥)
13-Aug-26	7,200	4,856
05-Aug-26	6,600	4,749
08-Jun-26	6,500	4,815
23-Mar-26	3,300	3,102
04-Feb-26	2,300	2,735
12-Jan-26	2,400	2,385
06-Nov-25	2,100	2,370
02-Oct-25	2,300	2,198
04-Aug-25	1,940	1,930
07-Jul-25	1,900	1,852
13-May-25	1,480	1,423
31-Mar-25	1,620	1,429
03-Feb-25	1,800	1,473
07-Nov-24	1,900	1,720
01-Oct-24	2,100	1,715
05-Aug-24	2,500	1,657
02-Jul-24	2,700	2,119
08-May-24	2,900	2,255
03-Apr-24	3,300	2,447
01-Feb-24	3,400	2,592
27-Nov-23	3,500	2,725
01-Nov-23	3,300	2,492
04-Oct-23	3,700	2,706

Murata Mfg. (6981.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	12,900	7,416
08-Jun-26	12,600	8,711
30-Apr-26	5,400	5,156
23-Mar-26	4,200	3,508
12-Jan-26	3,900	3,194
31-Oct-25	3,600	3,392
02-Oct-25	3,300	2,770
30-Apr-25	3,000	2,214
31-Mar-25	3,400	2,306
01-Oct-24	3,500	2,893
02-Jul-24	3,800	3,364
03-Apr-24	3,400	2,768
06-Dec-23	3,300	2,892
04-Oct-23	3,100	2,657

Ibiden (4062.T)

Date of report	Target price (¥)	Closing price (¥)
04-Aug-26	26,700	16,330
08-Jun-26	25,500	17,060
23-Mar-26	9,600	7,540
12-Jan-26	8,100	6,714
30-Oct-25	14,600	6,798
02-Oct-25	10,600	4,577
07-Jul-25	7,200	3,185
08-May-25	6,300	2,062
31-Mar-25	6,200	1,995
04-Feb-25	6,500	2,250

TDK (6762.T)

Date of report	Target price (¥)	Closing price (¥)
31-Jul-26	4,900	3,008
08-Jun-26	4,600	3,715
28-Apr-26	3,000	2,677
23-Mar-26	2,900	2,043
12-Jan-26	2,800	2,142
31-Oct-25	2,700	2,673
02-Oct-25	2,400	2,154
01-Aug-25	2,100	1,876
28-Apr-25	2,000	1,460
31-Mar-25	2,100	1,546

Ibiden (4062.T)

Date of report	Target price (¥)	Closing price (¥)
01-Oct-24	7,000	2,296
02-Jul-24	8,400	3,235
01-May-24	8,000	3,074
03-Apr-24	9,000	3,189
01-Feb-24	8,700	3,784
29-Nov-23	8,900	3,548
26-Oct-23	9,600	3,602

TDK (6762.T)

Date of report	Target price (¥)	Closing price (¥)
01-Nov-24	2,300	1,848
01-Oct-24	2,230	1,948
02-Sep-24	11,300	2,023
30-Jul-24	11,500	2,029
02-Jul-24	11,200	2,004
26-Apr-24	8,300	1,462
03-Apr-24	8,700	1,488
31-Jan-24	7,800	1,488
06-Dec-23	7,500	1,338
01-Nov-23	6,500	1,158
04-Oct-23	6,400	1,052

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures**Disclosures required by United States laws and regulations**

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; registered market maker. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors

should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock's total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region's Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts' investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs' involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, make a market or act as liquidity provider in, and buy or sell, any securities, derivatives, or other financial instruments referred to in this research (including any financial instruments issued by the companies referred to herein), unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.